LSU | College of Art + Design

# LA 7032 | Generative Landscapes

Brendan Harmon
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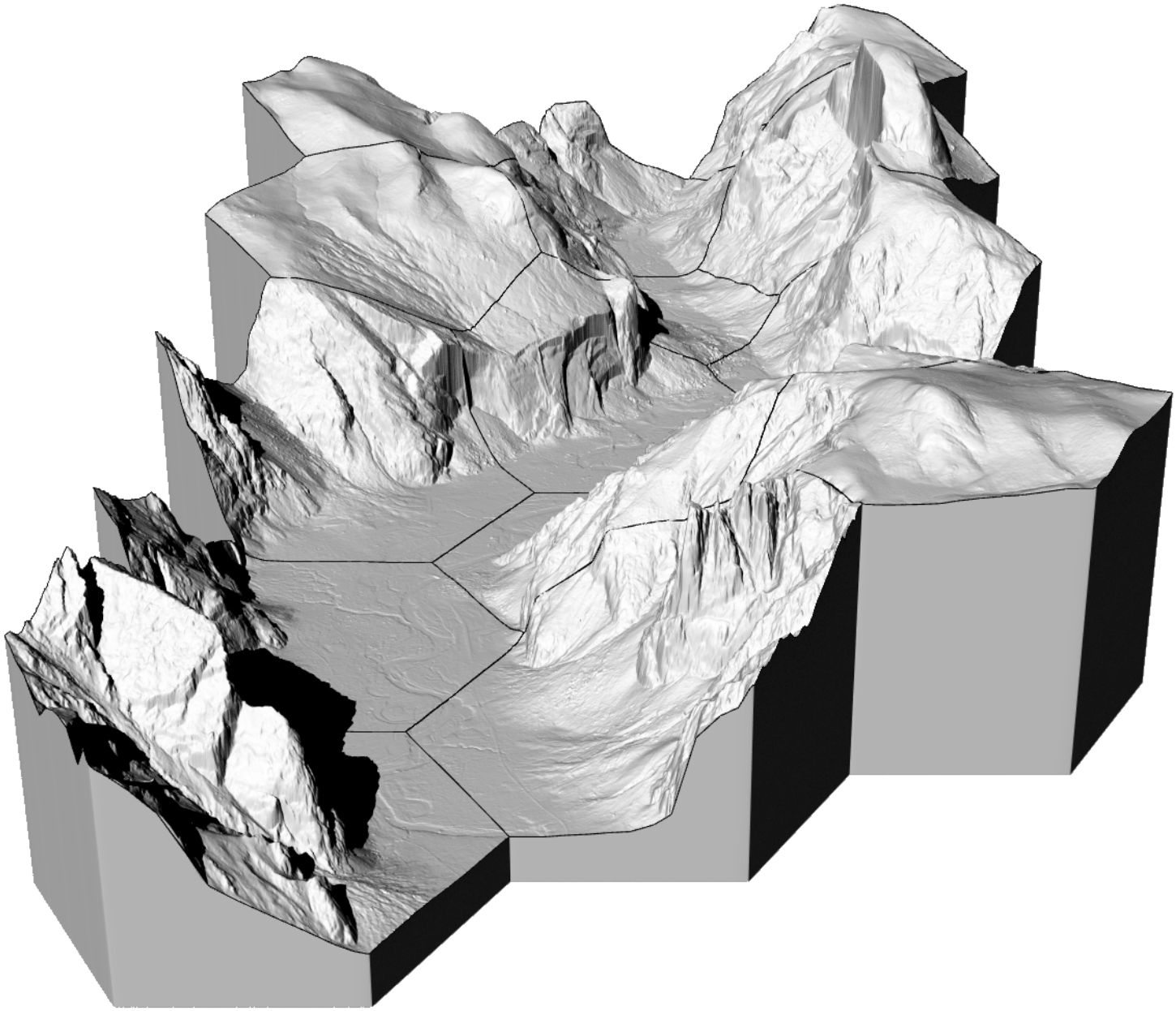
Spring 2020. Design 217.
Tuesday & Thursday 12:30am–3:20pm.



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## Course Description

This course is an introduction to generative design for landscape architects. In this course you will learn how to algorithmically model landscapes, numerically simulate physcial processes like the flow of water and sediment, and digitally fabricate landforms. You will conduct surveys with drones and terrestrial lidar, parametrically model variations on landfroms and planting using visual programming, photorealistically render your algorithmic planting patterns, and computer numerical control (CNC) mill your landforms. Through this course you will learn creative approaches to digital design and apply emerging technologies to landscape architecture.

### Keywords

- Algorithmic architecture
- Generative design
- Parametric modeling
- Visual programming
- Digital fabrication
- 3D rendering
- Lidar analytics
- Drone photogrammetry
- Geospatial programming

## Topics

### Lidar & drone analytics

1 Design week
2 Alphabet & algorithm
3 Drone surveying
4 Drone photogrammetry
5 Lidar analytics

### Generative design

6 Geospatial modeling
7 Geospatial programming
8 Generative design
9 Families of form
10 Parametric modeling

### Digital fabrication

11 Spring break
12 Attractors
13 3D rendering
14 CNC milling
15 Geospatial analytics

# Course Schedule

## Drones

| | | | |
|---|---|---|---|
| 01.14.2020 | **Studio** | Introduction | |
| 01.21.2020 | **Reading** | Alphabet & algorithmn | |
| 01.23.2020 | **Reading** | Digital fabrication in arch. | **Essay:** Generative design |
| 01.28.2020 | **Tutorial** | Intro to drones | |
| 01.30.2020 | **Fieldwork** | Drone survey | |
| 02.04.2020 | **Tutorial** | Drone photogrammetry I | |
| 02.06.2020 | **Tutorial** | Drone photogrammetry II | **Project:** Drone survey |
| 02.11.2020 | **Fieldwork** | Lidar survey | |
| 02.13.2020 | **Tutorial** | Point cloud processing | **Project:** Lidar survey |

## Generative

| | | | |
|---|---|---|---|
| 02.18.2020 | **Tutorial** | Terrain modeling | |
| 02.20.2020 | **Tutorial** | Hydrologic simulation | |
| 02.27.2020 | **Tutorial** | Geospatial programming I | |
| 03.03.2020 | **Tutorial** | Geospatial programming II | |
| 03.05.2020 | **Tutorial** | Generative design | |
| 03.10.2020 | **Tutorial** | Parametric patterns | |
| 03.12.2020 | **Tutorial** | Parametric landforms | **Project:** Families of form |

## Fabrication

| | | | |
|---|---|---|---|
| 03.17.2020 | **Tutorial** | Parametric planting | |
| 03.19.2020 | **Studio** | Workday | |
| 03.31.2020 | **Tutorial** | Attractors | |
| 04.02.2020 | **Tutorial** | Waveforms | |
| 04.07.2020 | **Tutorial** | CNC milling I | |
| 04.09.2020 | **Lab** | CNC milling II | |
| 04.14.2020 | **Tutorial** | 3D rendering I | |
| 04.16.2020 | **Tutorial** | 3D rendering II | |
| 04.21.2020 | **Tutorial** | Geospatial analytics I | |
| 04.23.2020 | **Tutorial** | Geospatial analytics II | **Project:** Parametric landscapes |
| 04.28.2020 | **Studio** | Workday | |
| 04.30.2020 | **Gallery** | Final review | |

## Projects

**Essay** Read Mario Carpo's *The Alphabet and Algorithm* and Nick Dunn's *Digital Fabrication in Architecture* and then in reponse write a 500-word critical essay. How have digital tools and processes transformed the practice of landscape architecture and how do you think they will shape the future of the discipline? How do you envision using digital design tools and processes in your work?

Dunn, Nick. 2012. *Digital Fabrication in Architecture*. Laurence King Publishing.

Carpo, Mario. 2011. *The alphabet and the algorithm*. Cambridge, MA: MIT Press.

**Drone survey** Conduct a topographic survey of the landform at Hilltop Arboretum with an unmanned aerial system (UAS) and automated ground control points. Use photogrammetry to generate a point cloud, digital surface model, and orthophoto. Upload your point cloud to Sketchfab.

**Lidar survey** Conduct a terrestrial lidar survey of the LSU College of Art & Design quad. Upload your point cloud to Sketchfab.

**Families of form** Use visual programming to simulate water and sediment flow for a series of nine landform variations in a virtual flume.

**Parametric landscapes** Use visual programming to parametrically model nine variations of landforms and planting. Photorealistically 3D render your variants with 3D trees, shrubs, and grasses. CNC mill models of your nine landforms in high density urethane foam. Upload your landform models to Sketchab.

Tedeschi, A. 2014. *AAD Algorithms-aided Design: Parametric Strategies Using Grasshopper*. Le Penseur.

## Software

Agisoft Metashape | https://www.agisoft.com/
CloudCompare | https://www.danielgm.net/cc/
Sketchfab | https://sketchfab.com/
GRASS GIS | https://grass.osgeo.org/
ArcGIS | https://www.esri.com/
Rhinoceros | https://www.rhino3d.com/
Grasshopper | http://grasshopper3d.com/
RhinoTerrain | http://www.rhinoterrain.com/
RhinoCAM | https://mecsoft.com/rhinocam-software/
Thea Render | https://www.thearender.com/

## Grading

| | | | |
|---|---|---|---|
| Essay | 5% | Familes of form | 20% |
| Drone survey | 20% | Parametric landscapes | 30% |
| Lidar survey | 20% | Course portfolio | 5% |

## Readings


Choma, J. 2015. *Morphing: A Guide to Mathematical Transformations for Architects and Designers*. Laurence King Publishing.

Stevens, J., and R. Nelson. 2015. *Digital Vernacular: Architectural Principles, Tools, and Processes*. EBL-Schweitzer. Taylor & Francis.

Tedeschi, A. 2014. *AAD Algorithms-aided Design: Parametric Strategies Using Grasshopper*. Le Penseur.

Beorkrem, C. 2013. *Material Strategies in Digital Fabrication*. Taylor & Francis.

Neteler, Markus, and Helena Mitasova. 2013. *Open source GIS: a GRASS GIS approach*. Vol. 689. Springer Science & Business Media.

Dunn, Nick. 2012. *Digital Fabrication in Architecture*. Laurence King Publishing.

Carpo, Mario. 2011. *The alphabet and the algorithm*. Cambridge, MA: MIT Press.

Deussen, Oliver, and Bernd Lintermann. 2010. *Digital Design of Nature: Computer Generated Plants and Organics*. Springer. doi:10.1007/b138606.

Picon, Antoine. 2010. *Digital culture in architecture: an introduction for the design professions*. 224. Boston, MA: Birkhaeuser.

Thompson, R. 2007. *Manufacturing Processes for Design Professionals*. Thames & Hudson.

Terzidis, Kostas. 2006. *Algorithmic architecture*. Elsevier Architectural Press.

Schodek, D., M. Bechthold, J.K. Griggs, K. Kao, and M. Steinberg. 2004. *Digital Design and Manufacturing: CAD/CAM Applications in Architecture and Design*. Wiley.

# Terminology

### Digital design

- Mass customization
- Generative design
- Parametric modeling
- Performative design
- Algorithm

### Spatial data

- Raster & Vector
- Array
- Point cloud
- Mesh
- Triangulated irregular network (TIN)
- Plain text
- Comma separated values (CSV)
- Integer & floating point numbers
- Quadtree & octree
- Non-uniform rational basis spline (NURBS)

### Geospatial

- Geographic information system (GIS)
- Digital terrain model (DTM)
- Digital elevation model (DEM)
- Digital surface model (DSM)
- Lidar
- Unmanned aerial system (UAS)
- Structure from motion (SfM)
- Delaunay triangulation
- Interpolation
- Regularized spline with tension (RST)
- Map algebra

### 3D rendering

- Ray tracing
- Diffuse shading
- Texture map
- Particle system
- Head mounted display (HMD)
- Cave automatic virtual environment (CAVE)

### Digital fabrication

- 3D printing
- Computer numeric control (CNC)
- Collet & Bit
- High density urethane (HDU)
- Medium density fiberboard (MDF)

## Policies

**Time Commitment Expectations** LSU's general policy states that for each credit hour, you (the student) should plan to spend at least two hours working on course related activities outside of class. Since this course is for three credit hours, you should expect to spend a minimum of six hours outside of class each week working on assignments for this course. For more information see: http://catalog.lsu.edu/content.php?catoid=12&navoid=822.

**LSU student code of conduct** The LSU student code of conduct explains student rights, excused absences, and what is expected of student behavior. Students are expected to understand this code: http://students.lsu.edu/saa/students/code.

**Disability Code** The University is committed to making reasonable efforts to assist individuals with disabilities in their efforts to avail themselves of services and programs offered by the University. To this end, Louisiana State University will provide reasonable accommodations for persons with documented qualifying disabilities. If you have a disability and feel you need accommodations in this course, you must present a letter to me from Disability Services in 115 Johnston Hall, indicating the existence of a disability and the suggested accommodations.

**Academic Integrity** According to section 10.1 of the LSU Code of Student Conduct, "A student may be charged with Academic Misconduct" for a variety of offenses, including the following: unauthorized copying, collusion, or collaboration; "falsifying" data or citations; "assisting someone in the commission or attempted commission of an offense"; and plagiarism, which is defined in section 10.1.H as a "lack of appropriate citation, or the unacknowledged inclusion of someone else's words, structure, ideas, or data; failure to identify a source, or the submission of essentially the same work for two assignments without permission of the instructor(s)."

**Plagiarism and Citation Method** Plagiarism is the "lack of appropriate citation, or the unacknowledged inclusion of someone else's words, structure, ideas, or data; failure to identify a source, or the submission of essentially the same work for two assignments without permission of the instructor(s)" (Sec. 10.1.H of the LSU Code of Student Conduct). As a student at LSU, it is your responsibility to refrain from plagiarizing the academic property of another and to utilize appropriate citation method for all coursework. In this class, it is recommended that you use Chicago Style author-date citations. Ignorance of the citation method is not an excuse for academic misconduct.